Fiscal Year:	FY 2015	**Task Last Updated:**	FY 01/03/2015
PI Name:	Shay, Jerry W. Ph.D.		
Project Title:	A Novel Biological Countermeasure and Mitigator of High LET-induced Cancer Progression		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-7208	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiation NSCOR/Virtual NSCOR NNJ10ZSA002N
Start Date:	01/01/2011	**End Date:**	12/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	4
No. of PhD Candidates:	5	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date is now 12/31/2014 per NSSC information (Ed., 1/22/14)		
Key Personnel Changes/Previous PI:	Ugur Eskiocak was a doctoral student who completed his PhD, Sept 2011. He is currently a postdoctoral fellow in the laboratory of Sean Morrison. Other students who were involved in specific aspects of this project (Peter Ly, completed his PhD, Nov 2012 and is currently in Don Cleveland's lab at the Ludwig Institute for Cancer Research at the University of California, San Diego, and Lu Zhang who completed her PhD in the Summer 2014 as a graduate student in the lab). Together with Aadil Kaisani (another graduate student, who completed his PhD May 2014, currently in medical school) and the addition of Mariam El-Ashmawy (MD/PhD graduate student who will complete her PhD in May 2015), we had a solid team of students and postdoctoral fellows (Dr. Sang Kim) assisting in this project. In addition, we have two recent trainees, a graduate student Krishna Luitel and a new diversity postdoctoral fellow, Ronald Bozeman.		
COI Name (Institution):			
Grant/Contract No.:	NNX11AC15G		
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Task Description:	This proposal addressed Radiation Carcinogenesis Cancer Specific Gap 8: Biological Countermeasures (BCM). While progress has been made since NCRP report 132 in 2000, we still do not have the data for assessing the need for BCMs. However, particle emission levels for fatal cancer risks are projected to be violated for a Mars mission under all possible scenarios. With current permissive exposure limits astronauts would not be able to go on deep space missions, such as to Mars and back. Thus, it is prudent to assume that even with new research to modify uncertainty reductions in fatal cancer risk projection models, that the development of BCMs should be actively investigated. The work completed was to investigate Bardoxylone methyl (BARD or CDDO), an oral available minimally toxic anti-oxidant inflammatory modulator that is currently in clinical trials for patients receiving radiotherapy for breast cancer to reduce dermatitis. CDDO is a synthetic triterpenoid and there is a significant molecular understanding of its mode of action. We have already completed investigations with low and high LET irradiation in both normal human lung and colonic epithelial cells and have shown that 10-50nM CDDO is an effective radiation BCM, reducing double-strand DNA damage and anchorage-independent cancer progression. Based on our results in human epithelial cells, we have also documented and published that CDDO is effective in reducing radiation increases in mouse models of cancer. Our results document that CDDO has an excellent chance of being developed into a potent BCM to protect against high LET irradiation. There are data showing that CDDO can protect against radiation-induced mucositis in a hamster cheek pouch assay. In addition, CDDO protects against radiation-induced intestinal damage in zebrafish development. Lastly, we have evidence that CDDO can mitigate gamma-irradiation (using the colony formation assay) if provided within 30 minutes after IR exposure. In summary, the main objective of this project to investigate if CDDO is a potent BCM and/or mitigator in mouse models that are predisposed to cancer has now been completed and the overall hypothesis confirmed. . Specific aim 1: To use a lung cancer susceptible mouse model that develops significantly more invasive lung tumors when high LET is provided in fractionated/protracted doses over 5 to 10 days (e.g., 0.2 Gy x 5 days or 0.1 Gy x 10 days 56Fe 1GeV, compared to single acute 1 Gy 56Fe). Our objective is to determine if pretreatment with CDDO prior to acute or fractionated IR reduces the number of invasive (e.g., more lethal) cancers. We will examine at least 3 different ions. Specific aim 2: To test CDDO as a BCM in mouse models exposed to SPE simulations. As part of the UT Southwestern Lung NSCOR and the Georgetown GI NSCOR, we have lung cancer and colon cancer susceptible mice that have been treated with solar particle simulations, single protons, and HZE ions. Specific aim 3: To test CDDO as a radiation mitigator by providing it to mice within 60 minutes after irradiation. We compared SPE simulations (total body exposure 1-2 Gy Protons) and at least two other ions (56Fe at 600 meV, LET 175 and 28Si at 300 MeV LET 70). The significance of the proposed work is that we demonstrated CDDO reduces cancer incidence and progression in two important tissues (lung and colon) that are susceptible to radiation-associated cancer using doses and doses rates recommended by NASA and likely to be experienced by astronauts on long term missions. This research provided new data to help reduce uncertainties in current risk projection models.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of our research suggest that CDDO should be pursued and developed into an oral available radiation countermeasure and mitigator that will not only be important to protect astronauts from the detrimental effects of space radiation, but may also have utility in protecting first responders to nuclear disasters and normal human tissues when radiotherapy is used to treat cancer patients.
Task Progress:	Specific Aim 1: To use the LA-1 lung cancer susceptible mouse model that develops significantly more invasive lung tumors when high-LET is provided in fractionated/protracted doses. Our objective is to determine if pretreatment with non toxic doses of CDDO-EA prior to acute or fractionated HZE irradiation reduces the number of invasive (e.g., more lethal) lung cancers. We will compare this to other priority HZE ions using fluences and energies that are representative of GCR including 28Si and 16O. We will also test a mouse model of colon cancer (CPC;APC). Specific Aim 2: To test CDDO-EA as a biological countermeasure in mouse models exposed to SPE simulations. Specific Aim 3: To test CDDO-EA as a radiation mitigator by providing it to mice within 30-60 minutes after irradiation (via IP and orally administered gavage). We will compare SPE simulations (total body exposure 2 Gy protons) and two other ions (56Fe at 600 MeV/n, LET 175 at dose ranges between 0.1 Gy and 0.5 Gy and 28Si at 350-600 MeV/n LET 70 at doses between 0.2 and 0.8 Gy). We will test single acute versus fractionated doses of 56Fe and 28Si. In summary, the lung cancer mouse model had the following conclusions. • 2Gy SPE and 1 Gy fractionated Fe- irradiation decreases the lifespan of LA1 K-ras animals and results in an increase incidence of carcinoma • LA1 K-ras mice on CDDO-EA diet during fractionated Fe- irradiation exhibit a decreased incidence in invasive carcinoma in comparison to mice on control diet • LA1 K-ras mice on CDDO-EA diet during solar particle irradiation simulations exhibit a decreased incidence in invasive carcinoma in comparison to mice on control diet • LA1 K-ras mice fed CDDO-EA without irradiation show reduced tumor progression • LA1 K-ras mice fed CDDO-EA prior to IR with lower doses of Fe-, 0.1 Gy-0.5 Gy) and Protons (single dose 50 and 150 MeV at 0.5-1 Gy total and SPE 0.5 Gy and 1.0 Gy with energies ranging from 50MeV to 150MeV). In summary, the colon cancer mouse model experiments had the following conclusions. • 2 Gy SPE - irradiation resulted in a 3-fold increase incidence of carcinoma • The CPC;APC mice on CDDO-EA diet during solar particle irradiation simulations exhibited fewer polyps ~100 days post –IR in comparison to mice on control diet • The CPC;APC mice on CDDO-EA diet during solar particle irradiation simulations exhibit a decreased incidence in

invasive carcinoma in comparison to mice on control diet ~250 days post-IR

• CPC;APC mice exposed to acute proton irradiation did not have as many invasive cancers compared to the SPE simulations. This suggests protracted proton exposure increases risks of cancer progression

• CDDO-EA, as a mitigator of high LET induced invasive cancers, was highly effective in reducing the incidence of invasive cancers

• We initiated mechanistic studies such as changes in circulating microRNA, and inflammatory signature post-IR with and without countermeasure protection.

Overall progress/conclusion:

1. The central focus of our proposed research was to provide the basis for improved estimation and uncertainty reduction for solid cancer from space IR and to develop evidence that biological countermeasures reduce the risk for solid cancer from HZE IR. The objective have been met and mostly published in peer reviewed journals.

2. We developed new approaches to biological countermeasures investigations for solid cancer risks for HZE nuclei.

3. We collaborated with NSCOR bioinformaticists to utilize systems biology approaches for linking several levels of biological organization. These new data provided information for the improved estimation and uncertainty reduction for high priority NASA ion beams. These included Protons, Silicon, and Iron using the recommended energies, doses, and dose-rates considered high priority by NASA.

4. We complemented and did not duplicate experiments already funded by NASA.

5. We tested specific mechanistically based hypotheses and not just collected more data.

6. We attended all required meetings and shared data, mice, tissues, DNA, and RNA as needed by other NASA funded investigators and published our results.

Bibliography Type:	Description: (Last Updated: 11/21/2025)
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